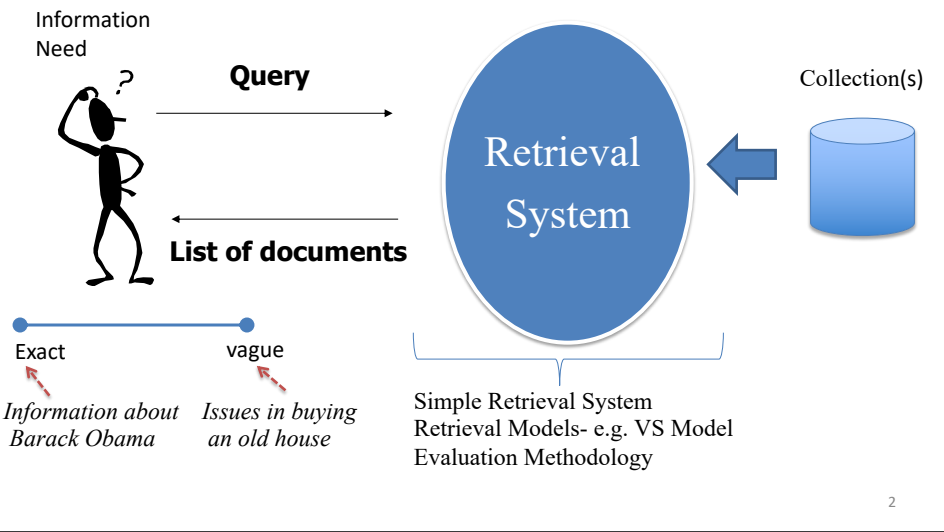


Relevance Feedback

Information Retrieval
Craig Macdonald
2026

1

Information Retrieval Thus Far



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Information Needs

- An *information need* is the underlying cause of the query that a person submits to a search engine
 - Sometimes called *information problem* to emphasise that information need is generally related to a **task**
- Categorised using a variety of dimensions
 - e.g., number of relevant documents being sought
 - Type of information that is needed
 - Type of task that led to the requirement for information

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Queries and Information Needs

- A query can represent very different information needs
 - May require different search techniques and ranking algorithms to produce the best rankings
- A query can be a poor **representation** of the *information need*
 - Users may find it difficult to express the information need
 - Users used to be encouraged to enter **short queries** both by the search engine interface, and by the fact that long queries often don't work
 - **Ambiguity**: the same query string may represent different information needs

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Query Ambiguity

- 16+% of user queries are ambiguous (Song et al., IP&M 2009)
 - e.g., what is a user issuing the query ‘ash’ after?



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Query Formulation Problem

- Difficult to generate well-formulated queries without
 - Knowledge of collection
 - How terms are distributed, type of docs, etc.
 - Retrieval environment
 - Term weighting, query language, retrieval strategy, etc
- First query is a trial run
 - Practically used to retrieve few useful items from a collection
 - Learn from those relevant ones
 - Query term modification

[Same is true even if an LLM is generating the query!]

Why is it difficult to formulate a query?

IR is an iterative process

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User Interaction

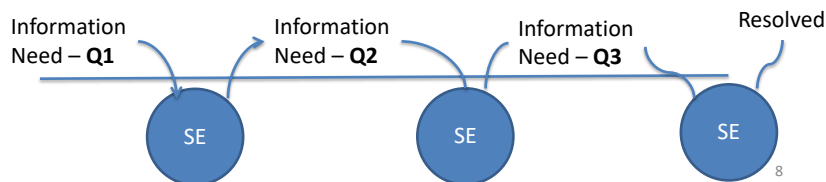
- **Interaction** with the system occurs
 - During query formulation and reformulation
 - While browsing the search results
- Key aspect of effective retrieval
 - Users can't change the **ranking algorithm** but can change the **results** through **interaction**
 - Helps refine the description of **information need**
 - e.g., same initial query, different information needs
 - How do users describe what they don't know?

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Interaction & Query Reformulation

- The user needs to find some information. Why?
 - To do a task
 - ‘**Anomalous**’ state of knowledge [Belkin 1982]
 - Maybe the user is uncertain about something
 - The user formulates a **query** (information need is transformed into a query)
 - System helps user to refine the query and ... accomplish the task



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ASK Hypothesis

- Belkin et al (1982) proposed a model called **Anomalous State of Knowledge**
- ASK hypothesis:
 - Difficult for people to define exactly what their information need is, because that information is a **gap** in their knowledge
 - Search engine should look for information that fills those gaps
- Interesting idea, little practical impact (yet)

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(Explicit) Interactions

“System helps user to refine the query”

Examples of **explicit interaction**:

- **Relevance Feedback**: the search engine permits the user to say what documents already retrieved are relevant or non-relevant
- **Query Expansion/Query Term Suggestion**: the search engine suggests additional query terms for the user
- **Query Suggestions**: the search engine suggests related queries for the user

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Relevance Feedback

- After initial retrieval results are presented, allow the user to provide **feedback** on the relevance of one or more of the retrieved documents
- Use this feedback information to **reformulate** the query
- Produce new results based on reformulated query
- Allows more **interactive**, multi-pass process

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Relevance Feedback Example

Permit user to select relevant documents

- ☐ 1. **Badmans Tropical Fish**
A freshwater aquarium page covering all aspects of the **tropical fish** hobby. ... to Badman's **Tropical Fish** ... world of aquariology with Badman's **Tropical Fish** ...
- ☐ 2. **Tropical Fish**
Notes on a few species and a gallery of photos of African cichlids.
- ☐ 3. **The Tropical Tank Homepage - Tropical Fish and Aquariums**
Info on **tropical fish** and **tropical** aquariums, large **fish** species index with ... Here you will find lots of information on **Tropical Fish** and Aquariums. ...
- ☐ 4. **Tropical Fish Centre**
Offers a range of aquarium products, advice on choosing species, feeding, and health care, and a discussion board.
- ☐ 5. **Tropical fish - Wikipedia, the free encyclopedia**
Tropical fish are popular aquarium **fish** , due to their often bright coloration. ... Practical Fishkeeping • **Tropical Fish** Hobbyist • Koi. Aquarium related companies. ...
- ☐ 6. **Tropical Fish Find**
Home page for **Tropical Fish** Internet Directory ... stores, forums, clubs, **fish** facts, **tropical fish** compatibility and aquarium ...
- ☒ 7. **Breeding tropical fish**
... intrested in keeping and/or breeding **Tropical**. Marine, Pond and Coldwater **fish** ... Breeding **Tropical Fish** ... breeding **tropical**, marine, coldwater & pond **fish** ...
- ☐ 8. **FishLore**
Includes **tropical** freshwater aquarium how-to guides, FAQs, **fish** profiles, articles, and forums.
- ☐ 9. **Cathy's Tropical Fish Keeping**
Information on setting up and maintaining a successful freshwater aquarium.
- ☐ 10. **Tropical Fish Place**
Tropical Fish information for your freshwater **fish** tank ... great amount of information about a great hobby, a freshwater **tropical fish** tank. ...

Top 10 documents for “tropical fish”

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Relevance Feedback Example

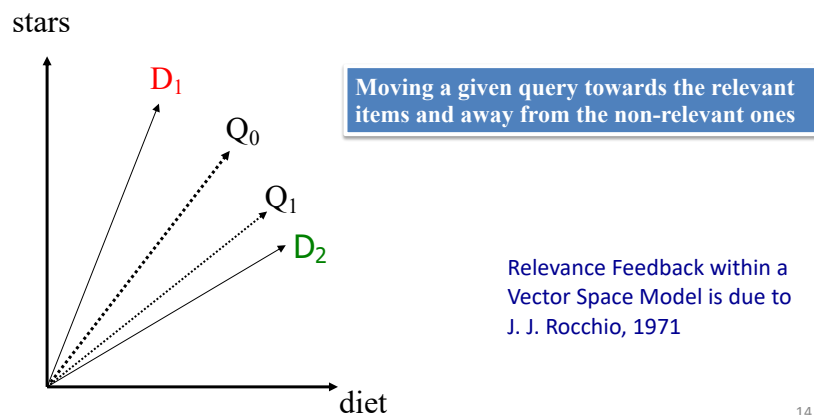
- Document 7 (“Breeding tropical fish”) has *explicitly* been indicated to be **relevant**
- In this document, the most frequent terms are:
breeding (4), fish (4), tropical (4), marine (2), pond (2), coldwater (2), keeping (1), interested (1)
- We can add these terms back into the query
 - This makes the query more **similar** to the relevant documents
 - E.g. {tropical fish} -> {tropical^5 fish^5 breeding^4 marine^2 ...}
- Specific weights and scoring methods used for relevance feedback depend on retrieval model

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Relevance Feedback Using Vectors

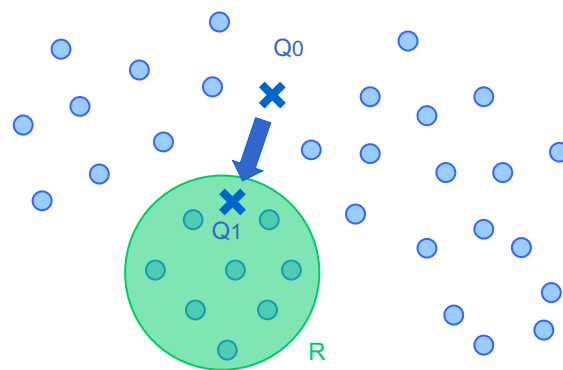
- Documents **relevant** to a particular query resemble each other in the sense that they are represented by **reasonably similar vectors**



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Conceptual View of Relevance Feedback



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Relevance Feedback

- Usually **both** of the following occur using the feedback:
 - Expand query with new terms
 - Re-weight terms in query
- There are many variations
 - Usually **positive** weights for terms from **relevant** docs
 - Found to be much valuable
 - Sometimes **negative** weights for terms from **non-relevant** docs
 - Remove terms that **ONLY** appear in **non-relevant** documents

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Query Reformulation for VSM

- Change query vector using vector algebra
- **Add** the vectors for the **relevant** documents to the query vector
- **Subtract** the vectors for the **non-relevant** docs from the query vector
- This **adds** both positive and negatively weighted terms to the query as well as **re-weighting** the initial terms

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Optimal Query

- Assume that the relevant set of documents C_r are known
- Then the **best query** that ranks all and only the relevant documents at the top is:

$$\vec{q}_{opt} = \frac{1}{|C_r|} \sum_{\forall \vec{d}_j \in C_r} \vec{d}_j - \frac{1}{N - |C_r|} \sum_{\forall \vec{d}_j \notin C_r} \vec{d}_j$$

where N is the total number of documents.

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Standard Rocchio Method

- Since all relevant documents are unknown, just use the **known** relevant (D_r) and non-relevant (D_n) sets of documents and include the initial query q_0

$$\vec{q}_{new} = \alpha \vec{q}_0 + \frac{\beta}{|D_r|} \sum_{\forall \vec{d}_j \in D_r} \vec{d}_j - \frac{\gamma}{|D_n|} \sum_{\forall \vec{d}_j \in D_n} \vec{d}_j$$

α : Tunable weight for initial query.

β : Tunable weight for relevant documents.

γ : Tunable weight for non-relevant documents.

Usually $\alpha=1$, $\beta=0.75$ and $\gamma=0.25$

Rocchio method vs optimal query?
What is the difference....
Why is it needed?

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Example: Rocchio Calculation

$D_{r1} = (.030, 0.00, 0.00, .025, .025, .050, 0.00, 0.00, .120)$ Relevant docs

$D_{r2} = (.020, .009, .020, .002, .050, .025, .100, .100, .120)$

$D_{n1} = (.030, .010, .020, 0.00, .005, .025, 0.00, .020, 0.00)$ Non-rel doc

$q_0 = (0.00, 0.00, 0.00, 0.00, .500, 0.00, .450, 0.00, .950)$ Original Query

$\alpha = 1$

$\beta = 0.75$

$\gamma = 0.25$

Constants

$$0 + (0.75 / 2) * (0.3 + 0.2) - (0.25 * 0.03) = 0.180$$

$q_{new} = \alpha q_0 + \left(\frac{\beta}{2} \times (D_{r1} + D_{r2}) \right) - \left(\frac{\gamma}{1} \times D_{n1} \right)$ Rocchio Calculation
Resulting Feedback Query

$q_{new} = (0.180, 0.000875, 0.002, 0.01, 0.527, 0.022, 0.488, 0.033, 1.04)$

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Other forms of Explicit Interaction (1)

Any CountryAny LanguageMatching any words

University of Glasgow

lilybankenquinflecttraininformjohnsondepartbarradelscurlquisquat

edinburghgroupsupportgardengogogcomput

2 ways to improve your search

1. tick appropriate words above and click Find

2. tick interesting documents below and click Find

Matches 1-10 of 1,152,670 matching your query

Individual word frequencies: univers: 1,125,289, glasgow: 51,902

Pegasus Workshop

Pegasus Workshop / Thursday 3rd and Friday 4th October 1996 / To be held in the University of Glasgow, Department of Computing Science, / 17 Lilybank Gardens, Glasgow G12 8RZ - Room 171 * Programme * Travel...

http://www.dcs.gla.ac.uk/~aleen/

Language: English Size: 6.8K Last modified: 1996-10-01

100% relevant, matching: univers glasgow

Author and publisher details

Contents Department Home Page / Guide to good practices for WWW authors / About the Guide to good practices for WWW authors / The Guide to good practices for WWW authors was written by Margaret Isaacs of...

http://www.dcs.gla.ac.uk/SIM/mmm.htm

Language: English Size: 3.1K Last modified: 1996-05-08

100% relevant, matching: univers glasgow

Searches related to information retrieval

information retrieval techniques

music information retrieval

information retrieval system

information retrieval research topics

information retrieval journal

information retrieval for music and motion

information retrieval book

information retrieval tools

Query (Term) Suggestions/ "Explicit" Query Expansion

Query Suggestions

Googooooooooooooole >

1 2 3 4 5 6 7 8 9 10 Next

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Other forms of Explicit Interaction (2)

information

information retrieval

information extraction

information retrieval definition

information retrieval techniques

information retrieval and data mining

information retrieval systems

information commissioner

information retrieval journal

information processing & management

information is beautiful

Information retrieval - Wikipedia, the free encyclopedia

https://en.wikipedia.org/wiki/Information_retrieval

More about Information retrieval

Query Auto-completions

Helping the user define the query on the fly

Similar Pages

Google™

sarah brightman

Search

Advanced Search

Preferences

Web Video Music

Sarah Brightman Official Website - Home Page

Official site of world's best-selling soprano. Join FAN AREA free to access exclusive perks, photo diaries, a global forum community and more...

www.sarah-brightman.com/ - 4k - Cache - Similar pages

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Relevance Feedback (RF) Performance

- RF generally improves retrieval performance (recall and precision)
- RF is most useful for **increasing recall** in situations where recall is important
 - Users can be expected to review results and to take time to iterate
- **Positive feedback** is more valuable than negative feedback
 - E.g. set $\gamma < \beta$; e.g. $\gamma = 0.25$, $\beta = 0.75$
 - Many systems only allow positive feedback ($\gamma=0$)

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Why is Relevance Feedback not Widely Used?

- Users are in general **reluctant** to provide **explicit** feedback (see Excite & Altavista experiments)
- Results in **long queries** that require more computation to retrieve
 - Search engines process lots of queries and allow little time for each one
- Makes it **harder** to understand why a particular document was retrieved

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Problems with (Explicit) Relevance Feedback (1)

- Relevance Feedback systems are recall-dependent
 - A small number of retrieved relevant docs can adversely affect the derivation of a new query
- Exploration-Exploitation trade-off
 - The balance between users visiting documents to assess relevance because *they want to* and because *they have to*
- Visiting documents to assess relevance is a **tedious, cumbersome** and **time-consuming** process

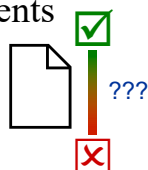
Relevance Feedback is a cognitively overloaded activity

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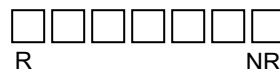
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Problems with Relevance Feedback (2)

- Two problems that are somewhat linked:
 - Treats **relevance** as a binary notion
 - Does not handle multi-topic or partially relevant documents



Query expansion formulae rely on collapsing complex relevance assessment scales into a binary notion

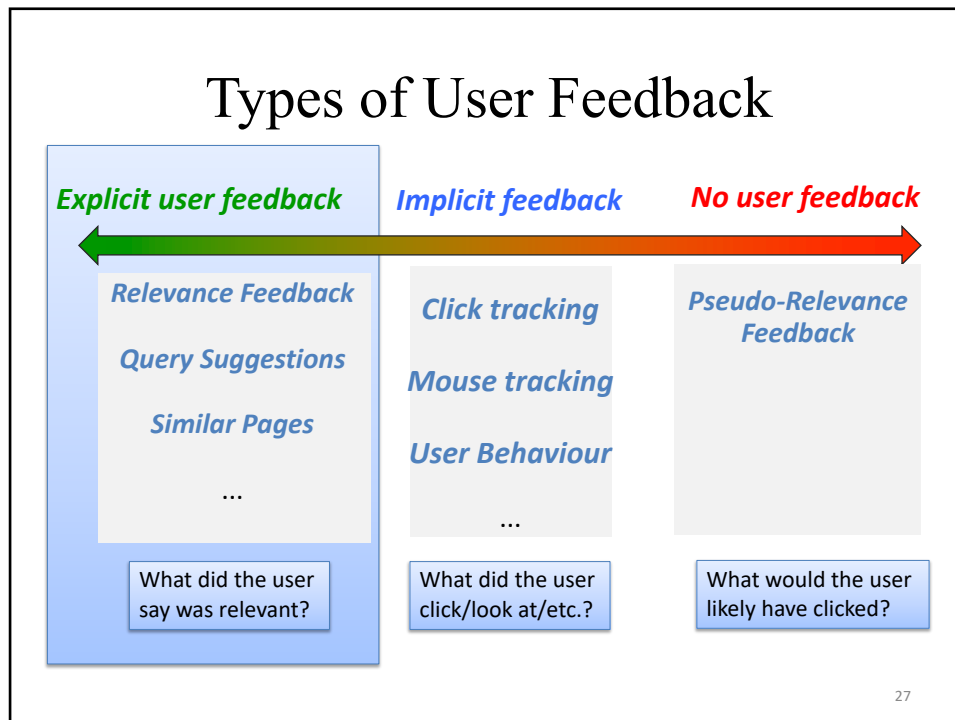


- Relevance is an abstract, intuitive concept that cannot be adequately expressed with 'yes/no'

Recall that user information need is vague and document representation is uncertain

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Implicit Relevance Feedback

- **Implicit Relevance Feedback** grew out of the **reluctance** of users to mark documents as **relevant** by clicking checkboxes next to document titles
 - Some searches are precision-oriented
 - Explicit feedback is **tedious**
- Can we gather **feedback** without requiring the user to do anything?
- **Idea:** gather **feedback** from observed **user behavior**
 - Attempting to determine what is **relevant** based on user **interaction** with a search system and returned documents

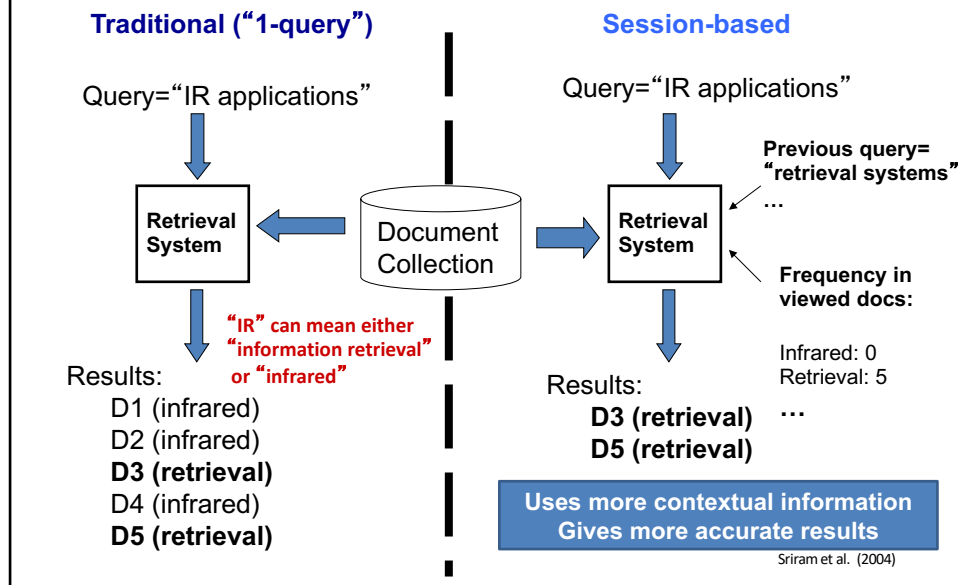
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Implicit Relevance Feedback in Search Engines

- Users no longer have to **click** checkboxes and browse to each document to assess it
- Typically, Web search engines **log users' interaction**
 - Clickthrough activity; Doc viewing time; Scrollbar activity, Mouse clicks; query reformulations; Search **session** activity
- Combine and use (**mine**) these to predict the user's intention and information need
- **Implicit Relevance Feedback** is **less accurate** than explicit RF
 - Confounding variables like *presentation bias*
 - Like much in Computing Science it is a trade-off!
 - But likely more useful than **pseudo-relevance feedback** (aka Blind RF), which contains no user judgments

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Traditional vs. Session-based Retrieval



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Pseudo-Relevance Feedback

- Use relevance feedback methods **without explicit user input**
- Just **assume** the **top m** retrieved documents are **relevant**, and use them to reformulate the query:
 - Look at the statistics of the terms in the top retrieved documents
 - Add the terms with highest weight to the query
 - Do relevance feedback (e.g. Rocchio) and carry out the retrieval with **new query**
- Aka (*automatic*) **Query Expansion (QE)**
 - A number of such QE models are implemented in Terrier

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Pseudo Relevance Feedback: Illustrative Example

- TREC Query: *Scottish highland games*
- Retrieve top 3 documents using the original query, and use the index to determine what other terms occur in those documents
- Using **Terrier's Bo1 QE** mechanism and Weak Stemming
 - Bo1 generalises Rocchio's approach by using a refined parameter-free statistical term weighting model (**no model parameters are needed**)
- The expanded query:
 - *Scottish highland games* Ligonier kilt caber clan toss Scot tartan **grandfather** **artist heavy** tradition dance Celtic dancer athlete heather competitor
- In the expanded query (using the relevance assessment)
 - These terms are helpful: Ligonier kilt caber clan toss Scot tartan
 - These terms bring noise: **grandfather** **artist heavy**
 - The rest of the added query terms are neutral, e.g. dancer, tradition

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Query Re-weighting

- Assume that in the expanded query, the terms are not re-weighted
- Assume that qtw is the frequency of the query term
- For the expanded query:
“Scottish highland games Ligonier kilt caber clan toss scot tartan tradition dance Celtic dancer athlete heather competitor grandfather artist heavy”
 - $qtw = 1$ for each query term
 - We can't differentiate the informative terms from the others
- Therefore, there is a need for **re-weighting** the query terms, including the expanded ones

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Term	Weight	Term	Weight
<i>Scottish</i>	<i>1.5000</i>	<i>highland</i>	<i>1.4087</i>
<i>games</i>	<i>1.3105</i>	<i>Ligonier</i>	<i>0.3609</i>
<i>kilt</i>	<i>0.2897</i>	<i>caber</i>	<i>0.1347</i>
<i>clan</i>	<i>0.1291</i>	<i>tradition</i>	<i>0.1189</i>
<i>dance</i>	<i>0.1115</i>	<i>Celtic</i>	<i>0.1067</i>
<i>toss</i>	<i>0.1062</i>	<i>dancer</i>	<i>0.1013</i>
<i>grandfather</i>	<i>0.1009</i>	<i>Scot</i>	<i>0.0895</i>
<i>athlete</i>	<i>0.0745</i>	<i>heather</i>	<i>0.0673</i>
<i>artist</i>	<i>0.0643</i>	<i>heavy</i>	<i>0.0606</i>
<i>tartan</i>	<i>0.0587</i>	<i>competitor</i>	<i>0.0427</i>

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Pseudo-Relevance Feedback Works

- Pseudo-relevance feedback **automates** the “manual” part of true (explicit) relevance feedback
- **Works well on average** in many experimental settings
- Example:
 - TREC 2005 Terabyte Track **ad hoc** task
 - Using Terrier’s **TF-IDF**, MAP=0.3024
 - Using Terrier’s **TF-IDF+Bo1**, MAP=0.3428
 - A **significant** improvement (p-value=0.008169)

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Problems with Pseudo-Relevance Feedback

- If the initial query is **hard**
 - That means the initial results are **poor**
 - Then pseudo-RF creates problems often drifting away from an optimal query
- May not be rewarding if the number of **relevant documents** is small
 - Adding more query terms cannot bring many relevant documents
 - e.g. query ‘*Homepage of Glasgow Information Retrieval Group*’ has only a **unique** relevant document. It is not helpful to expand the query

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Other Sources of Evidence for Query Expansion

- **Query Expansion** can examine different sources of evidence to automatically reformulate a query:
 - Top-ranked documents: based on **term co-occurrence** with query terms (**pseudo-relevance feedback**)
 - Also the entire corpus (incl. an external corpus), using a similar **term co-occurrence analysis**
 - Previous query reformulations of users (**query logs**)
 - Thesauri & dictionaries (e.g. **WordNet**) & **word embeddings**
 - However, they are not necessarily effective, as they do not take context (c.f. retrieved documents) into account

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Thesaurus

- A thesaurus provides information on **synonyms** and **semantically** related words and phrases.
- Example:

```
physician
  syn: ||croaker, doc, doctor, MD,
  medical, mediciner, medico, ||sawbones
  rel: medic, general practitioner,
  surgeon, ...
```

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Thesaurus-based Query Expansion

- For each term, t , in a query, expand the query with **synonyms** and **related words** of t from the thesaurus
- You might weight the **added** terms **less** than the original query terms
- Generally increases **recall**
- It could significantly **decrease precision**, particularly with ambiguous terms
 - “interest rate” → “interest rate fascinate evaluate”

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WordNet

- A more detailed database of **semantic relationships** between English words
- Developed by well-known cognitive psychologist George Miller and a team at Princeton University.
- About 155,000 English words.
- Nouns, adjectives, verbs, and adverbs grouped into about 175k **synonym sets** called **synsets**

<https://wordnet.princeton.edu/>

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WordNet Synset Relationships

- **Antonym**: front → back
- **Attribute**: benevolence → good (noun to adjective)
- **Pertainym**: alphabetical → alphabet (adjective to noun)
- **Similar**: unquestioning → absolute
- **Cause**: kill → die
- **Entailment**: breathe → inhale
- **Holonym**: chapter → text (part to whole)
- **Meronym**: computer → cpu (whole to part)
- **Hyponym**: plant → tree (specialization)
- **Hypernym**: apple → fruit (generalization)

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Query Expansion with WordNet

- The expanded query terms are usually **synonyms** of the original query terms (i.e. words in the same synset)
- Example: for “*Scottish highland games*”, we have the following expanded terms:
 - Scotch, Gaelic, upland, hilly, mountainous, **bet**, **gage**, **stake**, **punt** etc.
- It is also possible to add **hyponyms** (specialised terms) or to add **hypernyms** to generalise a query

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Statistical Thesaurus

- Existing human-developed thesauri are **not easily available** in all languages
 - Human thesauri are **limited** in the type and range of synonymy and semantic relations they represent
- Semantically related terms can be discovered from the **statistical analysis** of corpora such as term co-occurrence:
 - If term **t1** occurs in **X** documents, and a term **t2** occurs in **Y** documents and they co-occur in **C** documents:
 - Cosine (t1, t2) = $C/\sqrt{X*Y}$
 - Dice (t1, t2) = $2C/(X+Y)$
- Other approaches involve distributional semantics
 - **Word embeddings** - word2vec, glove, etc
- Do not usually work well compared to pseudo-relevance feedback

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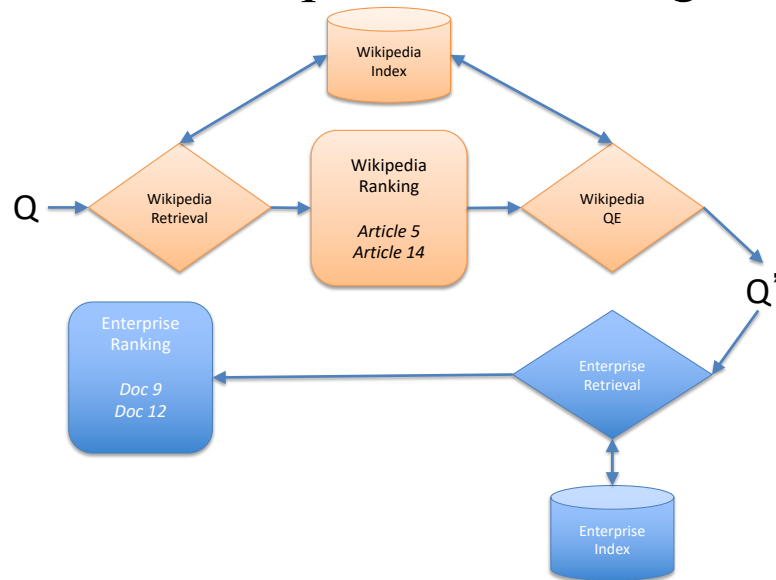
Query Expansion Using External Resource

- A **high-quality external** resource can bring useful information that improves the quality of the pseudo-relevance set
- [Kwok 2003]
 - The pseudo-relevance set contains top-ranked documents returned by Google!
 - Aka **Collection Enrichment** or **External Query Expansion**
- **Works well** on large collections, including Web & Microblog corpora
 - The larger the external resource is, the better the terms follow a random distribution
 - Advanced term weighting models are based on randomness statistical laws
 - A large external resource provides better collection statistics for the term weighting

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Example of External QE For Enterprise Search Engine



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Selective Query Expansion

- **Idea:** Disable query expansion if the pseudo-relevance set is predicted to be poor
 - Various **query performance predictors** have been proposed [He & Ounis, 2004]
 - For a given query, estimate how well a given query will do using a **statistical analysis** of query/corpus and/or the retrieved set of documents
 - E.g. are the query terms not informative enough, or is the query ambiguous (e.g. very topically dissimilar terms)
- **Works well**, and particularly appropriate for retrieval tasks that require early precision (e.g. Web search)

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Wisdom of the Crowds

- Consider the query: **Scotch whisky shop Bellevue**
- There are no such shops in Bellevue, so the user may reformulate to **Scotch whisky shop Seattle**
- If many users use similar query reformulations, from the **query logs** we can understand that **Bellevue** → **Seattle** is a possible reformulation/expansion:

Source	Target	Frequency/Score
bellevue	seattle	6
bellevue	bellevue ne	5
bellevue	bellevue washington	4

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Summary

- Relevance feedback is an effective means for user-directed query modification
- Modification can be done with either **direct** or **indirect** user input
- Modification can be done based on an individual's or a group's past input

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